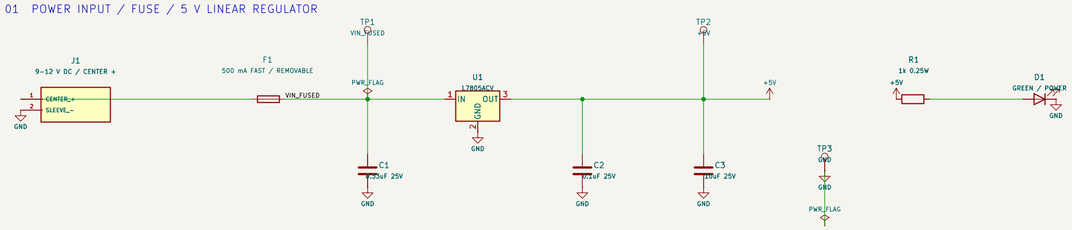
01 POWER INPUT / FUSE / 5 V LINEAR REGULATOR

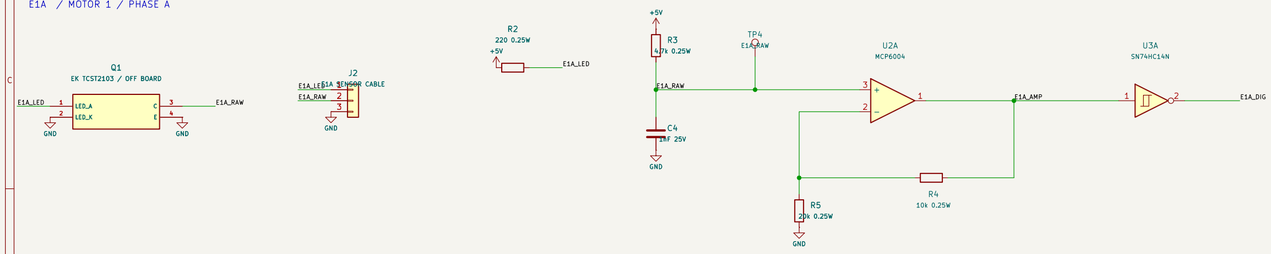


C1/C2 near U1: C3 bulk output. Heatsink required at high load.

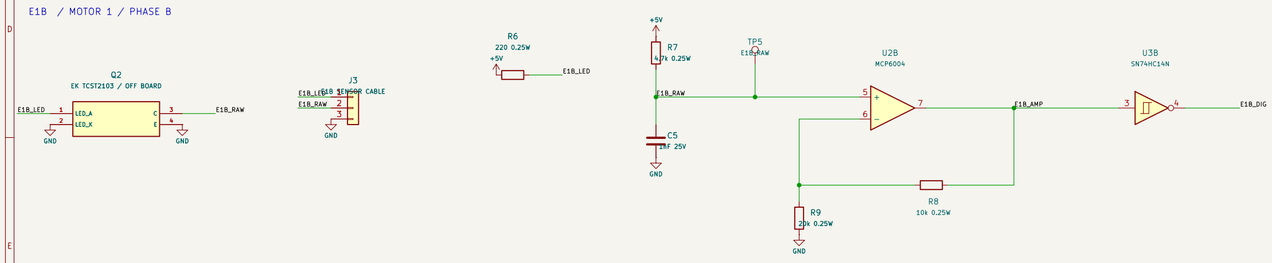
At 12 V, 250 mA: P(U1) = 1.75 W. Confirm load, fuse and thermal margin.

02 QUADRATURE OPTICAL SENSING / ACTIVE CONDITIONING

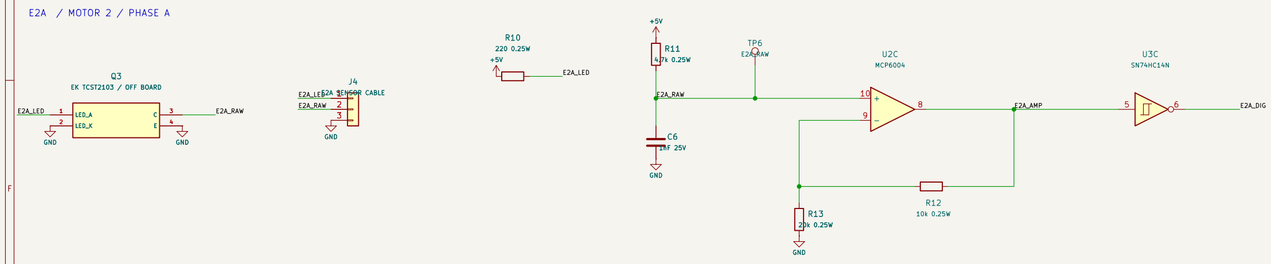
E1A / MOTOR 1 / PHASE A



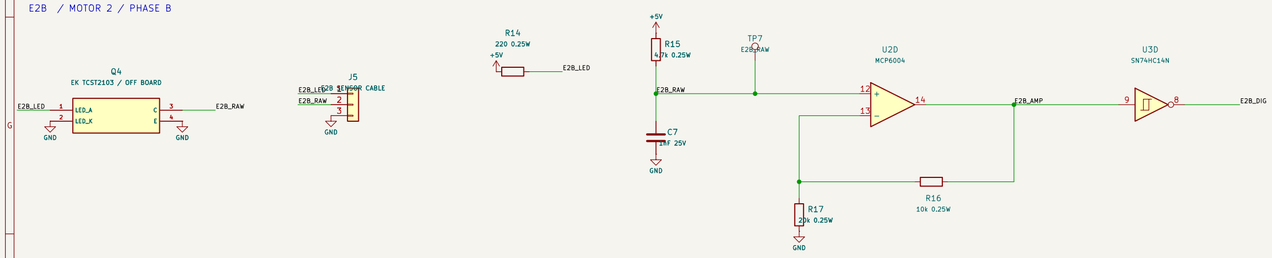
E1B / MOTOR 1 / PHASE B



E2A / MOTOR 2 / PHASE A



E2B / MOTOR 2 / PHASE B



DESIGN NOTES

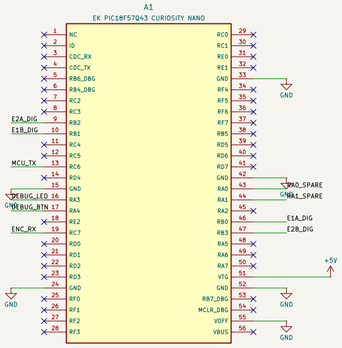
Four identical channels: gain = 1 + 10k/20k = 1.5; nominal RC pole = 33.9 kHz (4.7k / 1nF).

High input saturates U2 near +5 V; validate switching, recovery and maximum encoder frequency on the bench.

Two sensors per disk require 90 electrical-degree phase offset. Incremental position needs a home reference.

Sensor pin numbers are logical wiring IDs. Use Vishay D+/E+ markings; no sensor footprint assigned.

03 PIC18F57Q43 CURIOSITY NANO



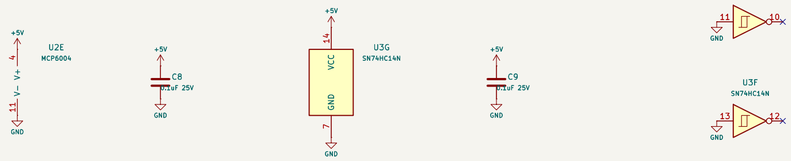
VTG 51 = +5 V; VOFF 55 = GND; VBUS 56 = NC.

Board edge numbers shown. Program RC6 TX / RC7 RX using PPS.

RB0..RB3 = encoder inputs. Unused pins intentionally NC.

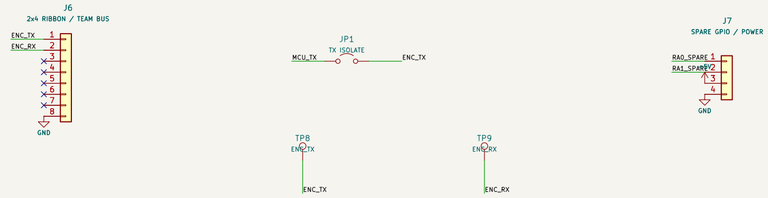
A1 represents the complete development board, not the bare MCU.

04 IC SUPPLY / UNUSED GATES



C8/C9/C10: 100 nF at each board/IC supply input.

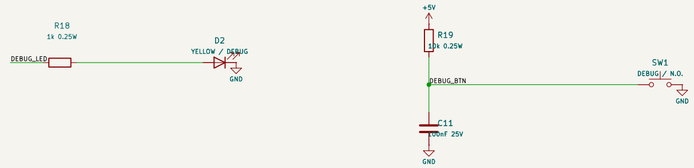
05 TEAM UART BUS / EXPANSION



J6 proposed: 1 TX to main RX; 2 RX from main TX; 8 common GND.

3–7 NC here. Confirm chain topology and numbering with team.

06 DEBUG LED / HARDWARE BUTTON FILTER



R19/C11 nominal 10 ms RC; enable Schmitt input and verify bounce.

Resistors 0.25 W; capacitors 25 V. Fit JP1 for normal operation.

No motor power on this PCB or ribbon cable.

ERC reviewed / bench and team interface validation pending

ASU RAS304 / Team 103 / Elijah Koiki

Sheet: /

File: Team103_Encoder_EK.kicad_sch

Title: Two-Motor Optical Encoder Subsystem

Size: A2	Date: 2026-10-27	**Rev: B**
KiCad E.D.A. 10.0.7		Id: 1/1